

MICHELE GRAFFEO

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RESEARCH INTERESTS

• Algebraic Geometry • Birational Geometry • Resolution of singularities • Hilbert schemes & Moduli spaces of sheaves • Representation theory • Toric Geometry • Enumerative Geometry • Minimal Model Program • Derived Category

ACADEMIC POSITIONS

Research Fellow RTDA at <i>Politecnico di Milano</i> - Milano (Italy)	3/2026-present
Postdoctoral Fellow at <i>SISSA</i> - Trieste (Italy)	1/2024-2/2026
Project: “ <i>Geometry of Hilbert schemes</i> ”	
Mentor: A. T. Ricolfi	
Postdoctoral Fellow at <i>Politecnico di Milano</i> - Milano (Italy)	2/2023-12/2023
Project: <i>PRIN 2020 “Squarefree Gröbner degenerations, special varieties and related topics”</i> (MUR, project number 2020355B8Y)	
Mentor: P. Lella	
Visiting Fellow at <i>SISSA</i> - Trieste (Italy)	1/2023-12/2023
Project: “ <i>Nested Hilbert schemes and GIT stability conditions</i> ”	
Mentor: U. Bruzzo	

ACADEMIC QUALIFICATIONS

Abilitazione Scientifica Nazionale – Professore di II Fascia , Settore concorsuale 01/A2 Geometria e Algebra, Italy	6/2026-present
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EDUCATION

PhD in Geometry and Mathematical Physics (cum Laude) at <i>SISSA</i> - Trieste (Italy)	10/2018-11/2022
Thesis: “ <i>Zero-dimensional sheaves, group actions and blowups</i> ”	
Supervisors: U. Bruzzo & A. T. Ricolfi	
Master of Science in Mathematics (cum Laude) at <i>University of Pisa</i> - Pisa (Italy)	9/2015-9/2018
Thesis: “ <i>Koszul cohomology and Hilbert schemes of points</i> ”	
Supervisors: M. Franciosi	
Bachelor in Mathematics at <i>University of Pisa</i> - Pisa (Italy)	9/2010-5/2015
Thesis: “ <i>Il teorema degli zeri in algebre analitiche reali e complesse</i> ”	
Supervisors: F. Acquistapace	
Scientific High School diploma at <i>Liceo Scientifico “Enrico Fermi”</i> - Siacca (Italy)	9/2003-8/2008

TO APPEAR AND PUBLISHED

• “ <i>The motive of the Hilbert scheme of points in all dimensions</i> ”, with S. Monavari, R. Moschetti and A. T. Ricolfi. To appear in <i>Proceedings of the London Mathematical Society</i>	2026
• “ <i>Classical Algebraic Geometry and Discrete Integrable Systems</i> ”, with G. Alecci and A. Stokes. To appear in <i>Symmetry and Integrability of Difference Equations</i> - Lecture notes of ASIDE15	2025
• “ <i>The geometry of double nested Hilbert schemes of points on curves</i> ”, with P. Lella, S. Monavari, A. T. Ricolfi and A. Sammartano. <i>Transactions of the American Mathematical Society</i> 378, 6013-6047 (2025)	2025
• “ <i>A counterexample to the parity conjecture</i> ”, with F. Giovenzana, L. Giovenzana and P. Lella. <i>Algebraic Geometry</i> , 12 (2025), no. 2, 173–188	2025
• “ <i>Unexpected but recurrent phenomena for Quot and Hilbert schemes of points</i> ”, with F. Giovenzana, L. Giovenzana and P. Lella. <i>Rendiconti del Seminario Matematico - Politecnico di Torino</i> , 82 (2024), no. 1, 145–170.	2024
• “ <i>5d Conformal Matter</i> ”, with M. De Marco, M. Del Zotto and A. Sangiovanni. <i>JHEP</i> , Volume 2024, article number 306, (2024)	2024
• “ <i>Moduli spaces of $\mathbb{Z}/k\mathbb{Z}$-constellations over $\mathbb{A}^2$</i> ”. <i>Communications in Contemporary Mathematics</i> 27 (2025), no. 03, 2450019	2024
• “ <i>Growth and integrability of some birational maps in dimension three</i> ”, with G. Gubbiotti. <i>Annales Henri Poincaré</i> , 13 July 2023	2023
• “ <i>On the Behrend function and the blowup of some fat points</i> ”, with A. T. Ricolfi. <i>Advances in Mathematics</i> , Volume 415, 15 February 2023, 108896	2023

PREPRINTS

- “Higgs Grassmannians”, with U. Bruzzo and B. Graña Otero. 2026
- “5d Trinions and Tetraons”, with M. De Marco, M. Del Zotto and A. Sangiovanni. 2026
- “Motivic and cohomological stabilisation of the Quot scheme of points”, with S. Monavari, R. Moschetti and A. T. Ricolfi. 2026
- “Components of the nested Hilbert scheme of few points”, with P. Lella. 2026
- “New components of Hilbert schemes of points and 2-step ideals”, with F. Giovenzana, L. Giovenzana and P. Lella. 2025
- “Invariants of nested Hilbert and Quot Schemes on surfaces”, with N. Fasola, D. Lewański and A. T. Ricolfi. 2025
- “Enumeration of partitions via socle reduction”, with S. Monavari, R. Moschetti and A. T. Ricolfi. 2025
- “The Painlevé equivalence problem for a constrained 3D system”, with G. Filipuk, G. Gubbiotti and A. Stokes. 2024

TEACHING

Lecturer

- (PhD course) *Computations in Algebraic Geometry* at SISSA - Trieste (Italy) 2026
- (PhD course) *Computations in Algebraic Geometry* at UFMG - Belo Horizonte (MG, Brazil) 2025
- (Advanced mini-course) *Classical Algebraic Geometry and Integrable Systems* at ASIDE - Milan (Italy) 2025
- (PhD course) *Complex algebraic surfaces* at SISSA - Trieste (Italy) 2025
- (PhD course) *Computations in Algebraic Geometry* at IMECC/UNICAMP - Campinas (SP, Brazil) 2025

Teaching Assistant

- (PhD course) *Topics in advanced algebra* at SISSA - Trieste (Italy) 2024-2025
- (PhD course) *Algebraic Geometry* at SISSA - Trieste (Italy) 2024-2025
- (PhD course) *Topics in advanced algebra* at SISSA - Trieste (Italy) 2023-2024
- (PhD course) *Algebraic Geometry* at SISSA - Trieste (Italy) 2023-2024
- Mathematical Analysis* at University of Trieste, School of Engineering - Trieste (Italy) 9/2023-2/2024
- (PhD course) *Algebraic Geometry* at SISSA - Trieste (Italy) 2022-2023
- Mathematical Analysis* at University of Trieste, School of Engineering - Trieste (Italy) 9/2022-2/2023
- Mathematical Analysis* at University of Trieste, School of Engineering - Trieste (Italy) 9/2021-2/2022
- Mathematical Analysis* at University of Trieste, School of Engineering - Trieste (Italy) 9/2020-2/2021
- Mathematical Analysis* at University of Trieste, School of Engineering - Trieste (Italy) 9/2019-2/2020
- Mathematical Analysis* at University of Pisa, School of Engineering - Pisa (Italy) 9/2017-2/2018
- Linear Algebra* at University of Pisa, School of Engineering - Pisa (Italy) 9/2017-2/2018
- Linear Algebra* at University of Pisa, School of Engineering - Pisa (Italy) 9/2016-2/2017

HELD SEMINARS, POSTER SESSIONS & WRITTEN ESSAYS

- “The geometry of the nested Hilbert schemes of points: components and their schematic structure” Politecnico di Milano
- “Super Geometry, a classical viewpoint” IGAP
- “The geometric approach to Integrable Systems: Spaces of initial conditions for Painlevé equations” SISSA
- “The geometric approach to Integrable Systems: Algebraic Entropy” SISSA
- “Elementary components of the nested Hilbert schemes of points & their schematic structure” UNICAMP Campinas (SP, Brazil)
- “A geometria do esquema de Hilbert aninhado” UFF Niterói (RJ, Brazil)
- “Crepan resolutions of canonical quotient singularities and moduli spaces of G-constellations” Belo Horizonte (MG, Brazil)
- “The Hilbert schemes of points: how to understand its geometry” Belo Horizonte (MG, Brazil)
- “The geometry of Hilbert schemes of points and 2-step ideals” Olinda (PE, Brazil)
- “The irreducible components of the Hilbert Scheme of points” ULB Bruxelles (Belgium)
- “A combinatorial approach to double nested Hilbert schemes of points” University of Pisa
- “The Hilbert scheme of points and its motive” João Pessoa (PB, Brazil)
- “New components of Hilbert schemes of points and 2-step ideals” UFF Niterói (RJ, Brazil)
- “Double nested Hilbert schemes of points on smooth curves” IMPA Rio de Janeiro (Brazil)
- “The geometry of the Hilbert scheme of points, and its nested variants” Razlog (Bulgaria)
- “La geometria dello schema di Hilbert di punti, e sue varianti” Isola delle Femmine (Palermo)
- “Irreducibility of the Hilbert scheme of points and the class of 2-step algebras” IME-USP São Paulo (Brazil)
- “Motives of the Hilbert schemes of points in all dimensions” ETH Zürich
- “Algebraic curves and one-dimensional complex manifolds” SISSA
- “The geometry of the Hilbert scheme of points and its variants” EPFL
- “Toric singularities” SISSA
- “The motive of the Hilbert scheme of points in all dimensions” University of Pisa
- Poster “Syzygies, Iarrobino’s example on 78 points and new components of Hilbert schemes” Jagiellonian University (Krakow)
- “Double nested Hilbert schemes of points” SISSA/IGAP
- “On the motives of the Hilbert schemes of points” University of Milan
- “Reducibility of $\text{Hilb}^{78}(\mathbb{A}^3)$ ” SISSA
- “Integrable systems and the Cremona-cubes group” University of Trieste
- “Nested variants of the Hilbert scheme of points” University of Milan

• “Nested variants of the Hilbert scheme of points on smooth curves”	SISSA
• “Double nested Hilbert schemes & reverse plane partitions”	Politecnico di Milano
• “Double nested Hilbert scheme of points on curves”	MIMUW
• “The geometry of double nested Hilbert schemes”	ETH Zürich
• “Some open problems and recent progress on the Hilbert schemes of points on smooth threefolds”	MPI MiS
• “The algebraic entropy and the Reye configuration”	TU Chemnitz
• “On the number twelve in algebraic geometry”	SISSA
• “On the dynamics of some birational maps of $\mathbb{P}^3$ ”	Politecnico di Milano
• “Behrend number and blowups of planar fat points”	Politecnico di Milano
• “Dynamics of some birational maps of the projective 3-space”	University of Genova
• “Dynamics of some birational maps of $\mathbb{P}^3$ ”	SISSA
• “GIT stability conditions on the space of G-Constellations”	University of Milan
• “Minimal resolutions of A_k singularities as moduli spaces of $\mathbb{Z}/(k+1)\mathbb{Z}$ -constellations”	Federal University of Paraíba
• Poster “Moduli spaces of $\mathbb{Z}/k\mathbb{Z}$ -constellations over $\mathbb{A}^2$ ”	SISSA
• “Moduli spaces of $\mathbb{Z}/k\mathbb{Z}$ -constellations over the affine plane”	University of Utrecht
• “On the Behrend function and the blowup of some fat points”	University of Bologna
• “How to get your hands dirty with canonical singularities”	SISSA
• “Crepan resolutions of symplectic quotient singularities as moduli spaces of constellations”	SISSA
• “Introduction to K3 surfaces”	SISSA
• “Moduli of representation of quivers and first examples of scattering diagrams”	SISSA/ICTP
• “Intersection theory and tautological ring of moduli space of curves”	SISSA
• “Blowups: some properties and funny examples”	SISSA
• “Towards the Kodaira vanishing theorem”	SISSA
• “Playing with quotient singularities”	SISSA
• “The real nullstellensatz”	University of Pisa
• “Normalization of complex spaces”	University of Pisa
• Fifty-pages extended essay on “Markov’s Theorem” based on in-class lectures and individual research	

ATTENDED SCHOOLS, WORKSHOPS & ADVANCED COURSES

• “Varieties with trivial canonical bundle and related geometry”	University of Milan	Spring 2026
• “MMNLP workshop 2026”	University of Milan	Winter 2026
• (Invited speaker) “Mini-workshop on flag supermanifolds, related supergeometries and applications”	IGAP (Trieste)	Winter 2026
• (Invited speaker) “Bandoleros 2025”	IMECC/UNICAMP (Campinas)	Fall 2025
• (Invited speaker) “LEGALzinho”	UFF (Niterói)	Fall 2025
• (Invited speaker) “First Brazilian Northeastern Meeting on Commutative Algebra and Algebraic Geometry”	(Olinda)	Fall 2025
• “TULSF X - A one-day algebraic geometry meeting”	(Ferrara)	Fall 2025
• (Invited speaker) “WAGP - Moduli Spaces in (Super)Geometry and Mathematical Physics”	(João Pessoa)	Fall 2025
• (Invited speaker) “Geometry in Algebra, Algebra in Geometry”	IMPA (Rio de Janeiro)	Fall 2025
• “Geometry In Bicocca”	Università di Milano-Bicocca	Summer 2025
• (Invited speaker) “WAGP - Geometry And Physics of Higgs Moduli”	(Razlog)	Summer 2025
• “Jordan Types of Artinian Algebras and Geometry of Punctual Hilbert Schemes”	Université Côte d’Azur	Summer 2025
• “Modules & Rings: Recent Developments in Commutative Algebra”	University of Genova	Spring 2025
• (Invited speaker) “Abecedarian of SIDE (ASIDE)”	University of Milan	Spring 2025
• (Invited speaker) “Giornate di Geometria Algebrica e Argomenti Correlati XVII”	Isola delle Femmine (Palermo)	Spring 2025
• “GC Legacy - A meeting in Algebraic Geometry”	Politecnico di Torino	Winter 2025
• “TULSF IX - A one-day algebraic geometry meeting”	SISSA (Trieste)	Fall 2024
• (Poster session) “Syzygies and Hilbert Schemes”	Jagiellonian University (Krakow)	Fall 2024
• (Invited speaker) “Algebro-geometric techniques for physics: bundles, stacks and supergeometry”	SISSA/IGAP	Spring 2024
• “The Geometry of Hilbert Schemes of Points”	CIRM - Levico Terme	Spring 2024
• (Invited speaker) “Genova-Torino-Milano Seminar”	University of Milan	Winter 2024
• “Enumerative geometry of the Hilbert scheme of points”	SRS Research Station (Les Diablerets)	Winter 2024
• “A day on Hilbert scheme of points”	Humboldt University (Berlin)	Fall 2023
• “Geometry In Bicocca”	Università di Milano-Bicocca	Summer 2023
• “A workshop on Geometry and Commutative algebra”	Politecnico di Milano	Summer 2023
• “Genova-Torino-Milano Seminar”	Università degli studi di Genova	Spring 2023
• “Hilbert schemes, moduli spaces, and symplectic varieties”	Université de Nantes	Spring 2023
• “Commutative Algebra TOwards Applications”	(Torino)	Spring 2023
• “Mini-school: Real and complex birational geometry”	at University of Milan (Milano)	Spring 2023
• “Refined invariants in Moduli Theory”	(Trieste)	Spring 2023
• (Invited speaker) “5th Christmas Workshop on Moduli Spaces and Integrable Systems”	(Genova)	Winter 2022
• “AGATES-Deformation theory workshop”	at IMPAN (Warsaw)	Winter 2022

• “Young Researchers Meeting in Algebra and Geometry 2022” conference at SISSA (Trieste)	Fall 2022
• “Recent Advances in Classical Algebraic Geometry” conference at Jagiellonian University (Krakow)	Summer 2022
• “Mini-workshop on Quiver Varieties and Related Topics” workshop at University of Oxford	Summer 2022
• (Poster session) “Integrable Probability, Classical and Quantum Integrability” workshop at SISSA	Spring 2022
• “New Perspectives on Hyperkähler Manifolds” workshop at Levico Terme	Spring 2022
• “Moduli Spaces and Stability Conditions” school & workshop at Levico Terme	Spring 2022
• “Derived Functors” PhD course by U. Bruzzo	Fall 2020
• “Hilbert schemes, McKay correspondence and singularities” winter school at Univ. Paris Diderot (Paris)	Winter 2019
• “Localisation in Enumerative Geometry” PhD course by A. T. Ricolfi	Fall 2019
• “Differentiable Orbifolds” PhD course by B. Fantechi	Fall 2019
• “Foliations in algebraic geometry” summer school at Istitut Fourier (Grenoble)	Summer 2019
• “Gauge Theory” PhD course by A. Tikhomirov	Spring 2019
• “Advanced topics in algebraic geometry” PhD course by E. Arbarello	Fall 2018
• “Algebraic surfaces: the cubic surface, the Cayley cubic, lines on smooth surfaces” PhD course by F. Catanese	Fall 2018
• “Cones of divisors and positivity” PhD course by L. Lombardi	Fall 2018
• “Integrable systems from moduli spaces of stable curves” PhD course by P. Rossi	Fall 2018

PRIZES & FUNDINGS

• (Principal Investigator) GNSAGA Research Project <i>Zero dimensional algebras and their moduli spaces</i> , Italy, 1,500 €	2026
• <i>Finanziamento Visite di studio e scientifiche in Brasile</i> , INDAM-IMPA, Italy 1.300 €.	2026
• (Workshop funding) <i>The geometry of Hilbert schemes of points</i> , GNSAGA-INDAM, Italy 2.500 €.	2024
• (Workshop funding) <i>The geometry of Hilbert schemes of points</i> , CIRM, Trento, Italia, 8.000 €.	2024
• <i>Lutman Prize</i> for the best PhD thesis SISSA in Mathematics	2023

MEMBERSHIPS

• GNSAGA-INDAM, Italy	Fall 2024-present
• INFN, Italy	Fall 2025-present

COMMITTEE MEMBER

• Committee member for master degree at EPFL, Switzerland	Winter 2025
• Committee member for PhD degree at Federal University of Paraíba, Brazil	Winter 2025

LANGUAGES & IT SKILLS

- Italian: native; English: fluent; Portuguese: basic; French: basic.
- Macaulay2, Sagemath, GAP, Latex, Unity, Windows OS, Android OS (Developer), Microsoft application, Office suite (ECDL) (Advanced), Ubuntu, C programming language, html.

REFEREES

Ugo Bruzzo	Paolo Lella	Andrea Tobia Ricolfi
SISSA	Politecnico di Milano	SISSA
bruzzo@sisssa.it	paolo.lella@polimi.it	aricolfi@sisssa.it

ORGANISATION OF EVENTS

• Co-organiser with U. Bruzzo of "Mini-workshop on flag supermanifolds, and supergeometries"	Trieste (Italy) Feb 26
• Co-organiser with U. Bruzzo, E. Pavia of the conference "TULSF - IX"	Trieste (Italy) Nov 24
• Co-organiser with U. Bruzzo, B. Graña Otero, D. H. Serrano, D. S. Gómez of the conference "WAGP24"	Trieste (Italy) June 24
• Co-organiser with P. Lella, S. Monavari, A. Ricolfi, A. Sammartano of the conference "GHiSP"	Levico Terme (Italy) May 24
• Co-organiser of the Algebraic Geometry seminar in SISSA	2021-22
• Co-organiser of the Algebraic Geometry seminar in SISSA/IGAP	2020-21
• Co-organiser of the Algebraic Geometry seminar joint between SISSA and ICTP	2019-20
• Co-organiser of the Algebraic Geometry seminar in SISSA	2018-19

OTHER TASKS

- Museum guide of a Mathematics exhibition named “*Mathematics in ancient Greece*” Pisa (Italy) 2018
- Developed strong analytical, problem-solving and time management skills, throughout my PhD studies at SISSA.
- Proven excellent communication, coaching and leadership skills, when working as a teaching assistant.
- Learnt how to be a team-player and how to get the best from joint outcome when working in a group.
- Learnt how to work and deliver results in high-pressure situations, such as studying and working at the same time.
- Volunteer work with both the needy and the elderly.
- Interests and hobbies: music, politics and chess.